
Software Requirements Specification

for

SmartMarket

Version 1.0 approved

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Organization: ---

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Revision History

Name	Date	Reason For Changes	Version
Abdulaziz Al-Ghamdi	23/10/2025	Initial creation and first approved version	1.0

1. Introduction

1.1 Purpose

This project aims to build a cloud-based, multi-tenant accounting system for grocery and supermarket retailers, featuring reliable credit (Ajel) management, and an e-commerce channel for orders with delivery integration.

1.2 Document Conventions

The document is bilingual (Arabic/English). Functional requirements use the tag REQ-X; non-functional use NFR-X.

1.3 Intended Audience and Reading Suggestions

Stakeholders include management, analysts, developers, QA, and operations. Read Section 2 (overview) first, then Sections 3–5.

1.4 Product Scope

A multi-tenant cloud accounting platform managing sales, purchases, inventory, customers, suppliers, credit, POS, reporting, and integrations with e-commerce and delivery apps.

1.5 References

- IEEE SRS Template (Karl Wiegers)
- Best practices for multi-tenant SaaS architecture

2. Overall Description

This section provides an overview of the SmartMart system and its intended environment.

2.1 Product Perspective

SaaS solution with a merchant dashboard and APIs for a separate customer-facing web/mobile app. Each store has isolated data and localized settings (tax, currency, language).

2.2 Product Functions

- Item & Inventory Management (barcodes, units, batches if needed)
- Sales & Invoicing + POS (tax invoice, discounts, returns)
- Customer & Credit (limits, schedules, alerts)
- Purchasing & Suppliers (POs, bills, adjustments)
- Treasury & Banking (receipts/payments, bank reconciliation)
- Reporting & Dashboards (sales, margin, stock movement, aging)
- Users & RBAC (roles, permissions, audit logs)
- E-commerce API integration, order & shipping management
- Delivery & Payments integrations
- Settings (configurable VAT/tax, currency, languages)

2.3 User Classes and Characteristics

- Store Owner/Manager — reports & configuration
- Accountant — journals, invoices, adjustments
- Cashier (POS) — sales operations
- Inventory Clerk — stock ops, cycle counts
- End Customer — e-commerce app/web

2.4 Operating Environment

Cloud-hosted (Web) with customer apps (Web/Mobile) via REST/GraphQL; relational DB, object storage, and POS/peripheral support.

2.5 Design and Implementation Constraints

- Multi-tenancy with data isolation.
- Local tax configurability.
- Arabic/English with RTL/LTR.
- Low-latency POS.
- Security, encryption, access governance.

2.6 User Documentation

- User Guide, POS quick guide, API docs, help center.

2.7 Assumptions and Dependencies

- Internet connectivity, certified payment/delivery, supported hardware.

3. External Interface Requirements

3.1 User Interfaces

- Merchant web dashboard (responsive, RTL).
- Streamlined POS with hotkeys.
- Notifications & alerts (email/SMS/in-app).

3.2 Hardware Interfaces

- Barcode scanners, thermal printers, cash drawers, scale (optional).

3.3 Software Interfaces

- REST/GraphQL for e-commerce/customer app.
- Payment gateways & delivery partners.
- Optional accounting integrations.

3.4 Communications Interfaces

- HTTPS/TLS, OAuth2/OIDC, webhooks, WebSocket (POS as needed).

4. System Features

SF-1 Items & Inventory

- REQ-1: CRUD for varieties, multiple barcodes, units of measurement.
- REQ-2: Track inventory and alarm thresholds.
- REQ-3: Inventory and Reconciliations with Audit Record.
- REQ-4: Price lists and offers.

SF-2 Sales, Invoicing & POS

- REQ-5: Invoices>Returns and Receipts/Disbursements.
- REQ-6: Configurable deductions and taxes.
- REQ-7: POS Offline Mode with Sync.
- REQ-8: Print POS receipts.

SF-3 Customers & Credit (Ajel)

- REQ-9: Files, Credit Limits, Payment Terms.
- REQ-10: Due Alerts and Notices.
- REQ-11: Debt Age Report and Automatic Stop When the Limit is Exceeded.

SF-4 Purchasing & Suppliers

- REQ-12: Purchase orders, supplier invoices.
- REQ-13: Quantities and Cost Adjustment.

SF-5 Treasury & Banking

- REQ-14: Receipts, Exchanges, and Transfers.
- REQ-15: Bank Reconciliation and Import of Statements.

SF-6 Reporting & Dashboards

- REQ-16: Sales and profit margin reports.
- REQ-17: Inventory Movement/Inventory Reports.
- REQ-18: Ages of Debt.

SF-7 Users & RBAC

- REQ-19: Precise roles and powers.
- REQ-20: Comprehensive Audit Log.

SF-8 E-commerce via API

- REQ-21: View items, prices, and inventory via API.
- REQ-22: Create and track the status of orders.
- REQ-23: Webhooks for updating payments/shipping.

SF-9 Delivery & Payments

- REQ-24: Integration of Delivery Providers (Shipping/Tracking/Costs).
- REQ-25: Multiple payment gateways and cash on delivery.

SF-10 Settings & Localization

- REQ-26: Configurable taxes for each store.
- REQ-27: Arabic/English and RTL/LTR.
- REQ-28: Currencies and Formats.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- NFR-1: $P_{95} \leq 300\text{ms}$ for web interfaces.
- NFR-2: POS Sale Flow $\leq 2\text{s}$; Sync $\leq 10\text{s}$.
- NFR-3: ≥ 1000 invoices/hour/store with horizontal expansion.

5.2 Safety Requirements

- NFR-4: Daily backup and geo-redundancy optional.
- NFR-5: Prohibition of the sale of prohibited/expired items.

5.3 Security Requirements

- NFR-6: TLS 1.2+ in transit and AES-256 for static data.
- NFR-7: OAuth2/OIDC , MFA , and Password Policies.
- NFR-8: Strict data isolation for each store.

5.4 Software Quality Attributes

- NFR-9: $99.9\% \geq$ available per month (excluding scheduled maintenance).
- NFR-10: Automatic tests $\geq 70\%$ for critical units.
- NFR-11: Horizontal expansion as the number of stores grows.
- NFR-12: Observability (Logs/Metrics/Tracing) with Alerts.

5.5 Business Rules

- BR-1: No exceeding the credit limit without authorization.
- BR-2: No invoice without payment method or approved term.
- BR-3: Returns are updated quantity and cost as per policy.

6. Other Requirements

- Internationalization & Accessibility (where applicable)
- Configurable local tax compliance.

Appendix A: Glossary

- Term (Ajel): Credit to the customer with a payment schedule.
- POS: Point of sale.
- RBAC: Role-based access control.

Appendix B: Analysis Models

- High-level DFD (TBD).

Appendix C: To Be Determined List

- Payment/Delivery Providers (TBD).
- Detailed Inventory Policies (TBD).
- SLA per package (TBD).